PHAN VAN BANG

AI Engineer — Multi-Agent Systems, RAG, Voice AI · Da Nang, Vietnam

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SUMMARY

AI Engineer with 3+ years shipping LLM products end to end (FastAPI, React, Docker, AWS): multi-agent systems, RAG, voice AI and IoT. Full-time at MekongAI (2023–2026), now freelancing. Also builds games, trading tools and AI video pipelines. Author of the open-source Nexus multi-agent platform (190+ MCP tools, 218 test files).

TECHNICAL SKILLS

AI / LLM	Multi-agent orchestration, RAG & GraphRAG (vector + full-text + graph, RRF), MCP (client & server, OAuth 2.1 PKCE), LangGraph, LangChain, LlamaIndex, CrewAI, LoRA/QLoRA fine-tuning (Unsloth), vLLM
Voice & Vision	Whisper STT, TTS (OpenAI, Fish Audio, ElevenLabs), VAD, PaddleOCR, SAM3, ComfyUI / Stable Diffusion, spaCy (vi/en)
Backend	Python, TypeScript, FastAPI, SQLAlchemy + Alembic, Celery, WebSocket, SSE, MQTT, REST
Data	PostgreSQL + pgvector, Qdrant, MongoDB, Redis, Neo4j
Frontend	React, Next.js, Zustand, Tailwind CSS
DevOps	Docker Compose, AWS (EC2, S3), GitHub Actions, Nginx, RunPod (GPU serverless), Pytest
Mobile / Games	Unity (C#, 2D/3D), React Native, Android (Java, Firebase), Flutter, C++ / OpenGL
Web	Laravel, Node.js / Express, Spring Boot (Servlet/JSP), Angular, Blockchain (Truffle, IPFS)
Trading	MQL5 Expert Advisors, Binance API, technical-analysis signal systems
Content / Media	AI video production (script, image, TTS, FFmpeg, CapCut), copywriting and brand-voice content, ComfyUI image workflows
Programming	Python, TypeScript/JavaScript, C#, Java, PHP, C/C++, Go

WORK EXPERIENCE

Freelance AI & Software Engineer

08/2026 – Present

- Build AI products for clients end to end: LangGraph diagnostic assistant for the automotive-repair domain (RAG, SSE streaming, S3 backup and restore) and YouTube video automation.
- Open-sourced Nexus (multi-agent platform) and published small dev tools (mcp-prune, sentinel-scan).

AI Engineer · MekongAI

06/2023 – 07/2026

- Architected **MekongAI Nexus**, a multi-agent orchestration platform (FastAPI, React 19, PostgreSQL + pgvector, Redis) with 190+ MCP tools and structured agent debate (challenge, rebuttal, synthesis).
- Built a hybrid RAG pipeline (vector + full-text + graph, RRF fusion) with GraphRAG memory; in internal evaluations it outperformed a vector-only baseline on retrieval accuracy.
- Designed layered agent memory (L0–L3), a continual-learning loop (skills, experiences, trajectory compression into SFT data) and a background memory-consolidation job.
- Hardened the runtime for production: circuit breaker and provider fallback across 8+ LLM backends, task checkpoint and auto-resume, command / SSRF / exfiltration guards, secret redaction, 218 automated test files.
- Shipped **Knowlympic**: PaddleOCR pipeline plus LLM question generation that cut content-creation time for 3 English language centers (roughly 80% by internal estimate).
- Built the **MetaHuman** voice platform (Whisper STT, streaming TTS, VAD) with sub-second voice response; connected ESP32 voice bots to the same backend via MCP and MQTT.
- Delivered computer-vision and GenAI services: 20+ ComfyUI workflows, a face-similarity API, SAM3 and PaddleOCR-VL on RunPod GPU serverless.
- Fine-tuned Llama 3.1 (8B/70B) with Unsloth (LoRA/QLoRA) on trajectory-compressed SFT data; delivered 50+ projects across Legal, Education, Recruitment, POS, Healthcare, Crypto, Social Media and Content Generation.

SELECTED PROJECTS

MekongAI Nexus – open-source multi-agent platform github.com/pvbang/nexus

- 25 REST endpoint modules, 68 MCP tool modules, OpenAI-compatible API, SSE streaming, 9 chat channels (Telegram, Discord, Slack, WhatsApp, Email, ...).
- Docker Compose deployment, 19 Alembic migrations, React 19 UI with live process cards for agent reasoning, debate and tool calls.

  Python 3.12, FastAPI, React 19, TypeScript, PostgreSQL + pgvector, Redis, LiteLLM, Docker

AI Vehicle-Repair Assistant (client project)

- LangGraph agent with a 5-layer pipeline (reasoning, search, evaluation, answer, verification) answering multi-DTC diagnostic queries with cited references, streamed over SSE.
- One-command bootstrap for the full stack; MongoDB backup to S3 with SHA-256 verification and staging-first restore.

  LangGraph, FastAPI, MongoDB, Qdrant, Redis, Streamlit, AWS S3, Docker

AI Recruitment Platform – Vietnamese NLP matching

- CV parsing with OCR and a dual spaCy pipeline (vi + en); job-candidate matching by vector similarity.

  FastAPI, Qdrant, spaCy, OCR, AWS S3

M-Auto – AI YouTube video studio

- Idea-to-upload pipeline: LLM scripts, image and TTS generation, FFmpeg rendering (Ken Burns, xfade, karaoke captions) with an AI quality critic that re-runs weak stages.

  Python, Streamlit, FFmpeg, Edge-TTS, faster-whisper

B. Nexus Quantum – MetaTrader 5 trading EA (commercial) b-nexus-quantum.vercel.app

- 11-module MQL5 Expert Advisor: 21 entry modes, 18-layer weighted signal scoring, multi-layer risk controls, licensing; sold with a public landing page.

Open-source developer tools github.com/pvbang

- mcp-prune: strips cosmetic fields from MCP tool schemas, cutting tokens by 73.8% on the bundled fixture while validating the same inputs.
- sentinel-scan: zero-dependency static scanner for JS/TS (hardcoded secrets, eval/exec sinks, weak crypto).

GAMES, MOBILE & WEB PROJECTS

- **Eternal Odyssey**: Unity 3D action RPG built in a team of two (5 playable classes, weapon tiers, EXP and PvE/PvP design); plus a Unity 2D platformer and a C++/OpenGL racing game.
- **Android chat app** (Java, Firebase, Jitsi Meet): messaging, image/video sharing, voice and video calls, voice search; 5 stars on GitHub.
- **React Native apps**: Vocab Master (English learning with OpenAI/Google AI), TransVoiceHub (speech-to-text, translation, text-to-speech), a math-solver app and "The Escape" game with a Node.js API.
- **Web apps**: Laravel shoe shop and restaurant management systems, Java client-server meeting app (screen share, chat), Angular + Truffle + IPFS blockchain demo.
- **Chrome extension** and **Capture2Text**: select or capture text on screen, then summarize, translate or answer with an LLM, with results copied to the clipboard.

EDUCATION & LANGUAGES

B.Eng. in Information Technology · Vietnam-Korea University of Information and Communication Technology (VKU), Da Nang 2020 – 2025

- Languages: Vietnamese (native), English (professional working proficiency: technical reading and writing).